

The Early Detection Of Neurodegenerative Diseases (EDoN) Initiative: Update On Cohort Profiles, Data Harmonisation, And Next Steps

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EDoN
Early Detection of
Neurodegenerative
diseases



BACKGROUND & METHODS

The Early Detection of Neurodegenerative diseases (EDoN) Initiative is investigating the potential utility of **low-burden digital measures** to support the early detection of dementia at scale.



EDoN is made up of 4 partner cohorts:

- CODEC-II** = Predictors of COgnitive DEcline using digital devices
- BU** = Boston University Alzheimer's Disease Research Center Health Outreach Program for the Elderly cohort
- WAMS** = Western Australia Memory Study
- HBA** = Healthy Brain Ageing Program

EDoN participants (including patients with SCD or MCI, and healthy controls) used a minimum of 2 of the following **4 digital measures**:

Mezurio smartphone app
Actively assesses cognition:

- Memory
- Exec. Function
- Motor control
- Speech

Longevity smartphone app
Passively assesses routine:

- Typing
- Scrolling
- Text input
- Screen time

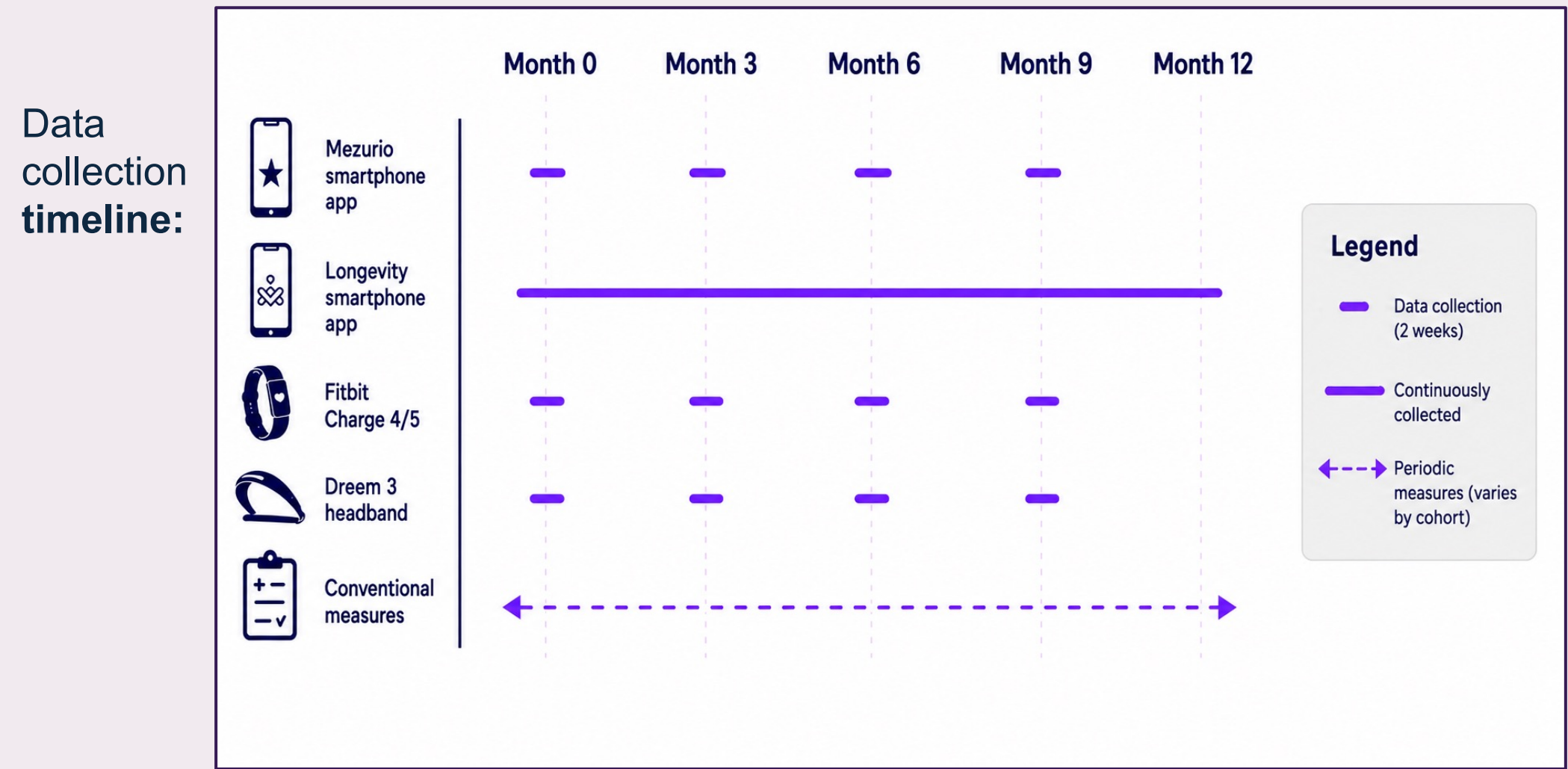
Fitbit Charge 4/5
Activity tracker capturing:

- Heart rate
- Phys. activity
- SpO2
- Sleep quality

Dreem 3 headband
Sleep EEG measuring:

- Movement
- Breathing
- Pulse rate
- Sleep stages

The EDoN cohorts also capture a range of **conventional measures** according to their existing protocols, including:



COHORT PROFILES

	CODEC-II (N = 135)	BU (N = 138)	HBA (N = 89)	WAMS (N = 70)	TOTAL (N = 432)
Age (yrs)	68.2 (9.5)	72.5 (10.0)	68.0 (7.8)	70.8 (7.7)	70.0 (9.3)
Sex (female)	61 (45%)	87 (63%)	67 (75%)	46 (66%)	261 (60%)
Education (yrs)	13.0 (3.3)*	17.3 (2.6)	15.0 (2.8)	14.2 (2.5)	15.0 (3.3)
Ethnicity†					
White	125 (93%)	116 (84%)			241 (88.3%)†
Chinese	3 (2%)	0 (0%)			3 (1.1%)†
Black	2 (1%)	20 (15%)			22 (8.1%)†
Asian	2 (1%)	1 (1%)			3 (1.1%)†
Any other	3 (2%)	1 (1%)			4 (1.5%)†
Diagnosis‡					
HC	25 (19%)	101 (73%)	10 (11%)	22 (31%)	158 (37%)
SCD	49 (36%)	4 (3%)	56 (63%)	44 (63%)	153 (35%)
MCI	56 (41%)	31 (22%)	23 (26%)	4 (6%)	114 (26%)
Dementia	5 (4%)	2 (1%)	0 (0%)	0 (0%)	7 (2%)

Continuous data are mean (SD). Categorical data are N (%). HC = Healthy control; SCD = Subjective cognitive decline; MCI = Mild cognitive impairment. *Data for 12 participants were missing. †HBA and WAMS did not collect ethnicity data; as a result, the 'Total' percentages for 'Ethnicity' reflect proportions in CODEC-II and BU only. ‡A total of 29 participants had an (additional) diagnosis of rapid eye movement sleep behaviour disorder (RBD). The number of participants with RBD in each cohort are: CODEC-II (N = 3); BU (N = 1); and HBA (N = 25); there were no individuals with RBD in WAMS.

DIGITAL DATA



EDoN participants used a minimum of 2 of 4 digital measures. Of 432, 378 (88%) selected the active **Mezurio smartphone app**, 343 (79%) selected the passive **Longevity smartphone app**, 366 (85%) selected the **Fitbit Charge 4/5**, and 313 (72%) selected the **Dreem 3 headband**.

MEASURE	HC (N = 158)	SCD (N = 153)	MCI (N = 114)	Dementia (N = 7)
Mezurio app				
Participants	128 (81%)	152 (99%)	93 (82%)	5 (71%)
Longevity app				
Participants	117 (74%)	133 (87%)	87 (76%)	6 (86%)
Fitbit Charge 4/5				
Participants	118 (75%)	149 (97%)	93 (82%)	6 (86%)
Dreem 3 headband				
Participants	91 (58%)	142 (93%)	77 (68%)	3 (43%)

Participant data is N (%). In this note we briefly describe the missingness and compliance data for the devices and apps, using the latest available figures at the time of finalising this poster. Compliance data was only available for Mezurio as of June 2026. Mezurio compliance was calculated as the % of scheduled task instances completed during active study periods. Compliance was >80% for HC, SCD and MCI, and >70% for Dementia. Missingness data was calculated by examining the proportion of protocol-required 'visits' (e.g., Months 0, 3, 6, 9 and – for Longevity – 12) missed across participants who agreed to use the app/device in question (calculated for each diagnostic category separately). Missingness was <10% across all apps/devices. HC = Healthy control; SCD = Subjective cognitive decline; MCI = Mild cognitive impairment.

COGNITIVE DATA



N (%) participants with conventional cognitive data available:

DOMAIN	CODEC-II (N = 135)	BU (N = 138)	HBA (N = 89)	WAMS (N = 70)
Global	110 (81%)	120 (87%)	89 (100%)	66 (94%)
Executive function	110 (81%)	119 (86%)	89 (100%)	67 (96%)
Psychomotor speed	110 (81%)	120 (87%)	89 (100%)	67 (96%)
Attention	110 (81%)	120 (87%)	89 (100%)	67 (96%)
Working memory	0 (0%)	120 (87%)	89 (100%)	67 (96%)
Episodic memory	110 (81%)	120 (87%)	89 (100%)	67 (96%)
Verbal fluency	110 (81%)	120 (87%)	89 (100%)	67 (96%)
Naming	110 (81%)	120 (87%)	89 (100%)	0 (0%)
Visuospatial	110 (81%)	119 (86%)	89 (100%)	67 (96%)
Premorbid IQ	105 (78%)	0 (0%)	88 (99%)	12 (17%)

Below we present the specific tests used to measure each domain across cohorts (n cohorts including a given test):

Global: CAMCOG-R (1), MoCA (2), MMSE (4)
Executive function: TMT-B (4), D-KEFS (1)
Psychomotor speed: TMT-A (4), WAIS DSST (1)
Attention: Digit/Number Span Forwards (3), Logical Memory Test Immediate (3)
Working memory: Digit/Number Span Backwards (3)
Episodic memory: Logical Memory Test Delayed (3), Craft Story 21 (1), RAVLT (2), CVLT II (1)
Verbal fluency: Category fluency–Various (4), Letter fluency–Various (4)
Naming: ADAS-Cog Naming Objects and Fingers (1), MINT (1), BNT (2)
Visuo-spatial/constructional: CAMCOG-R Visuospatial subscore (1), Benson Figure (1), RCFT (2), Clock drawing (1)
Premorbid IQ: TOPFUK (1), WTAR (1), CCRT (1)

NEUROIMAGING DATA



N (%) participants with neuroimaging data available:

MODALITY	CODEC-II (N = 135)	BU (N = 138)	HBA (N = 89)	WAMS (N = 70)
MRI				
T1-weighted	103 (76%)	53 (38%)	86 (97%)	54 (77%)
T2/FLAIR*	103 (76%)	53 (38%)	86 (97%)	54 (77%)
DTI	0 (0%)	53 (38%)	19 (21%)	0 (0%)
fMRI	0 (0%)	53 (38%)	19 (21%)	0 (0%)
QSM	0 (0%)	0 (0%)	22 (25%)	54 (77%)
PET				
Amyloid	8 (6%)	6 (4%)	2 (2%)	56 (80%)
Tau	0 (0%)	8 (6%)	0 (0%)	0 (0%)

DTI = Diffusion tensor imaging; FLAIR = Fluid attenuated inversion recovery; fMRI = Functional magnetic resonance imaging; MRI = Magnetic resonance imaging; PET = Positron emission tomography; QSM = Quantitative susceptibility mapping. *T2 and FLAIR grouped for poster-level summary; some cohorts obtained both T2-weighted and FLAIR images.

BLOOD SAMPLE DATA



N (%) participants with blood, plasma, and/or serum sample(s) available:

	CODEC-II (N = 135)	BU (N = 138)	HBA (N = 89)	WAMS (N = 70)
N with sample(s) available	111 (82%)	113 (82%)	85 (96%)	69 (99%)

PROGRESS

- ☒ Of 432 individuals enrolled, 382 (88%) completed 12 months of digital data collection
- ☒ Two published qualitative papers explored EDoN participants' perspectives of how acceptable the digital measures were (scan the QR codes for the articles!)
- ☒ The study protocol manuscript is under review at a peer-reviewed journal
- ☒ UCL data experts (led by Dr Neil Oxtoby) are wrangling EDoN data (the data available is summarised on this poster)
- ☒ EDoN is bringing together the cohort PIs and their teams for our first in-person Investigators' meeting during **AAIC 26**



NEXT STEPS



Cognitive, clinical and neuroimaging data will be **harmonised**

Data will be **shared** via a cloud-based secure research environment

Evaluate the potential utility of digital measures to facilitate the **early detection** of dementia